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Saveetha Institute of Medical And Technical Sciences
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Dynamic Public Parking Pricing System Based on Real-Time Demand Prediction

CSA0925-PROGRAMING WITH JAVA

Research Scholar

D. Chand basha(192424218)

Ganeshram. M(192321060)

Monish. S(192311132)

Saveetha School of Engineering

Supervisor

Dr. Sivasankar C

Dept of CSE

Saveetha School of Engineering

Problem Background

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Application Domain

- Smart City Transportation and Intelligent Parking Management.
- AI- and data-driven urban mobility solutions.

Current Scenario

- High-demand areas become congested while other parking spaces remain underutilized.
- Drivers spend significant time searching for available parking.

Need for the Project

- Introduce dynamic pricing based on real-time demand prediction.
- Improve parking space utilization and city traffic management.

Target Users

- Private vehicle owners.
- Parking facility operators.

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Problem Statement

Problem Definition

- Develop a system that predicts real-time parking demand and dynamically adjusts parking prices to optimize parking utilization and reduce urban congestion.

Evidence / Statistics

- Drivers spend **15–30%** of urban travel time searching for parking.
- Fixed pricing often leads to overcrowded parking facilities during peak hours.

Limitations of Current Systems

- Static pricing does not adapt to changing demand.
- Inefficient allocation of parking spaces.
- Limited support for smart city parking management.

Expected Impact

- Reduced traffic congestion and parking search time.
- Increased parking revenue and support for sustainable smart cities

Project Objectives

Primary Goal: To develop a Dynamic Public Parking Pricing System that predicts real-time parking demand and dynamically adjusts parking prices.

1

Objective 1

[e.g., Achieve ≥90% demand prediction accuracy using historical and real-time occupancy data]

2

Objective 2

[e.g., Reduce average parking search time by ≥25% within the target zone]

3

Objective 3

[e.g., Deploy a responsive user notification system with latency under 5 seconds]

Scope: The project focuses on implementing a dynamic parking pricing system that uses real-time and historical parking data to predict demand and adjust parking fees accordingly.

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Literature Survey

S.No	Year	Author(s)	Title	Key Contribution / Findings	Research Gap / Limitation
1	**2022**	Ali & Shah	**IoT-Based Smart Parking Monitoring System**	Developed an IoT-enabled parking monitoring system using occupancy sensors and cloud connectivity for real-time parking availability.	Does not support dynamic pricing or parking demand prediction.
2	**2023**	Zhang et al.	**Machine Learning-Based Parking Demand Prediction**	Applied machine learning algorithms to forecast parking demand using historical occupancy and traffic data.	Prediction accuracy decreases during special events and unexpected traffic conditions.
3	**2023**	Zhou & Lee	**Dynamic Parking Pricing Using Demand–Supply Analysis**	Proposed a demand-based pricing model to optimize parking occupancy and improve revenue.	Relies mainly on historical data; ignores real-time weather and event information.
4	**2024**	Sarker et al.	**Smart Parking Management System with IoT Integration**	Integrated IoT sensors, mobile applications, and cloud services for real-time parking management and monitoring.	High deployment cost and limited scalability for large metropolitan cities.
5	**2024**	Mabrouk et al.	**Deep Learning-Based Parking Demand Forecasting**	Utilized deep learning models to predict parking demand and recommend optimal pricing strategies.	Requires large datasets and high computational resources for training.

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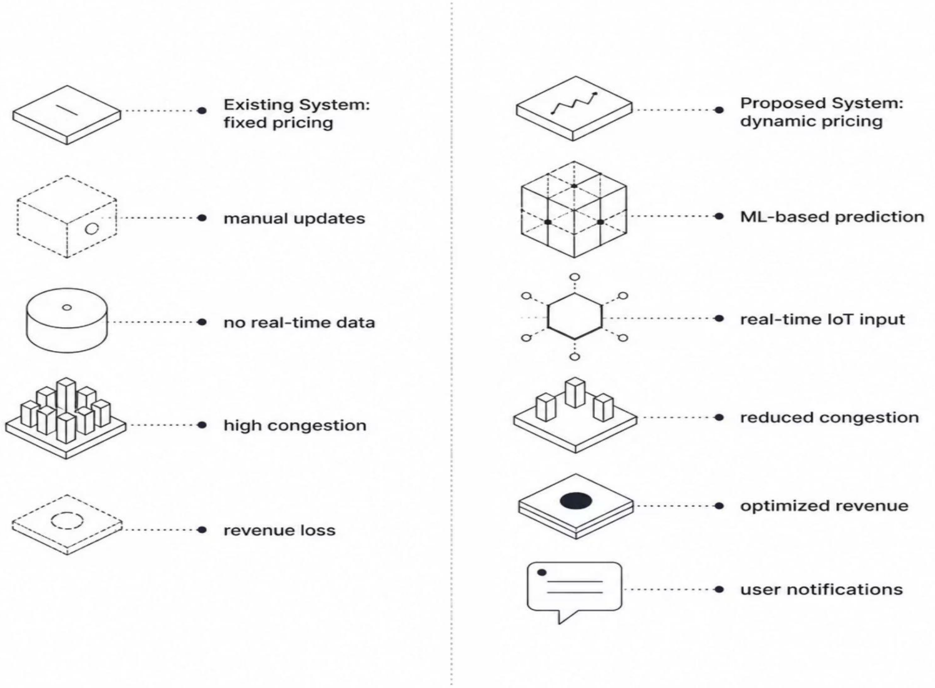
Existing vs. Proposed System

Current Approach:

Traditional public parking systems use fixed pricing without considering real-time demand, leading to inefficient parking utilization and increased traffic congestion.



Novelty:

Uses real-time parking demand prediction instead of static pricing. Dynamically adjusts parking fees based on occupancy, demand, and time.

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System Requirements

Functional Requirements

- Real-time demand prediction from live sensor feeds
- Automated dynamic price adjustment per zone/lot
- User notifications via mobile app / display boards

Non-Functional Requirements

- Scalable architecture supporting N+ parking zones
- System latency < 5 seconds end-to-end
- 99.5% uptime SLA for production deployment

Hardware & Software Stack

- **Hardware:** [Sensors, edge devices, servers, IoT gateways]
- **ML Frameworks:** [TensorFlow, scikit-learn, PyTorch]
- **Databases:** [PostgreSQL, InfluxDB, Redis]
- **Data Sources:** [Occupancy feeds, traffic APIs, weather data]

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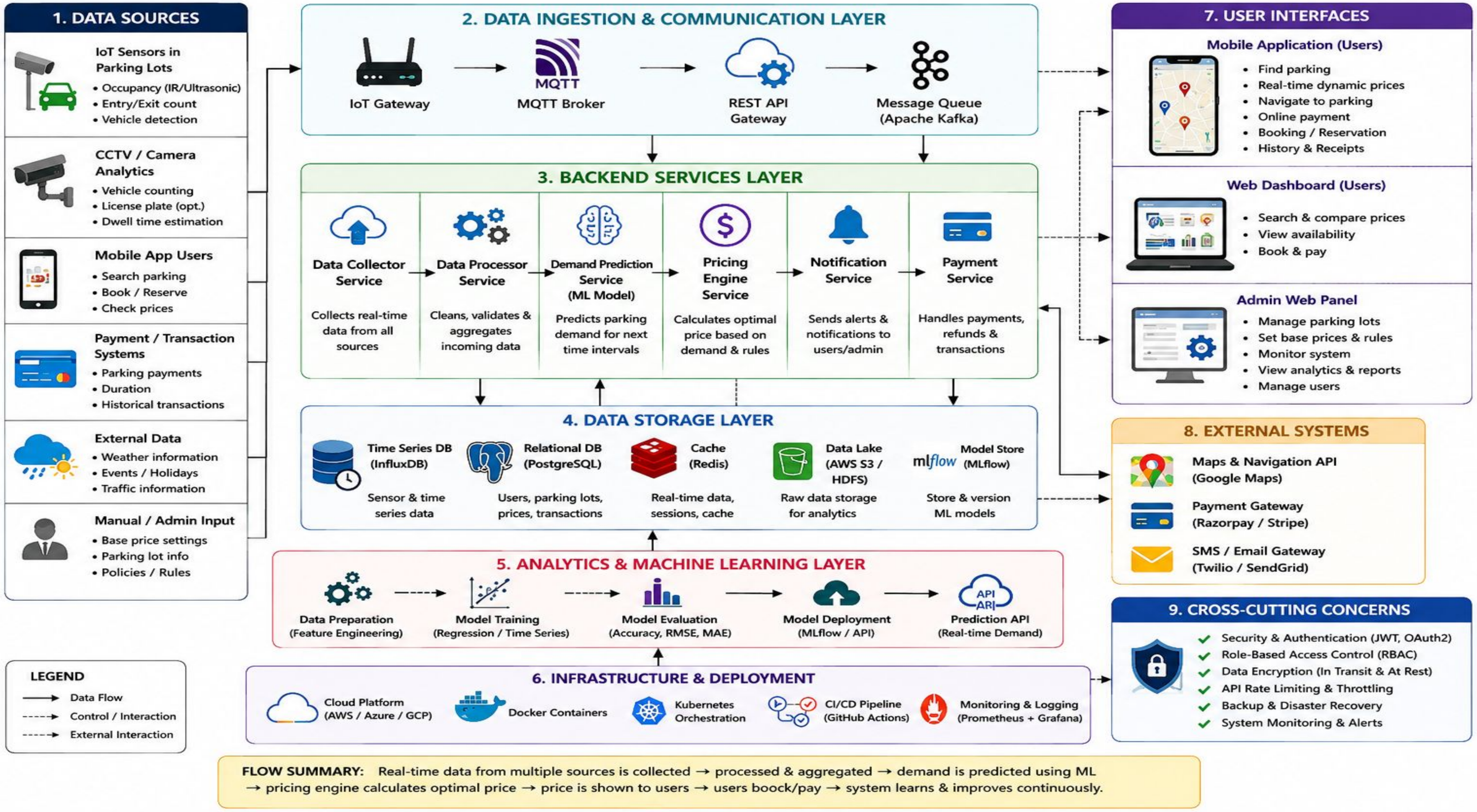
System Architecture



Project Planning

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Work Breakdown Structure

01

Requirements & Research
[Timeline — Weeks 1–3]

02

System Design
[Timeline — Weeks 4–6]

03

Development & ML Training
[Timeline — Weeks 7–14]

04

Testing & Validation
[Timeline — Weeks 15–18]

05

Deployment & Documentation
[Timeline — Weeks 19–22]

Milestones:

Key Milestones & Risks

- ✅ Prototype ready — [Target date]
- ✅ ML model trained and validated — [Target date]
- ✅ Full system deployed — [Target date]

Key Risks:

- **Data Availability:** Sensor coverage gaps or incomplete historical datasets
- **Model Accuracy:** Overfitting to specific zones; generalization challenges
- **User Adoption:** Resistance from drivers or operators to dynamic pricing

Expected Outcomes & Impact

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Deliverables

- Fully functional dynamic pricing system
- Trained and validated ML demand prediction model
- Research paper and technical documentation

Benefits

- Reduced parking search time for drivers
- Optimized parking revenue for city authorities
- Lower vehicle emissions from reduced cruising

Future Work

- Multi-city and multi-zone deployment at scale
- Integration with autonomous vehicle routing systems
- [Additional extensions — EV charging, public transit linkage]

thank you!